

Identifying Variables — Set A — Answer Key

Level 3 · Ages 11-14 · Answers in bold. Accept any answer that means the same.

Independent variable — the factor you change, and the values you test.

Dependent variable — the factor you measure, with its units.

Controlled variables — factors kept the same in every trial.

Control group — a separate no-treatment trial used for comparison. If there isn't one, write "none".

- ① Twenty bean seedlings grow in identical pots of the same potting soil on one windowsill, each given 100 mL of water a day. Each week, groups of five get 0, 5, 10 or 15 drops of liquid fertilizer. After four weeks, every plant's height is measured.

Independent variable (and values):

Drops of fertilizer per week: 0, 5, 10 and 15.

Dependent variable (and units): **Height after four weeks, cm (or mean height per group).**

Three controlled variables: **Any three: identical pots; same potting soil; 100 mL of water a day; same windowsill (light and temperature); four weeks.**

Control group: **The five plants given no fertilizer (0 drops).**

- ② Students take their pulse for 15 seconds after 2 minutes each of sitting, walking and jogging, resting until it is back to the sitting value in between. Each count $\times 4$ gives beats per minute. (Anyone who should not jog can record.)

Independent variable (and values):

Type of activity: sitting, walking, jogging.

Dependent variable (and units): **Heart rate (beats per minute).**

Three controlled variables: **Any three: 2 minutes per activity; 15-second count $\times 4$; rest back to the sitting pulse; the same students.**

Control group: **None — no separate group (accept: sitting as the baseline).**

- ③ Twenty radish seeds are placed on damp paper towel in each of four identical dishes with lids, kept at 5 °C, 15 °C, 25 °C and 35 °C. Each dish gets 10 mL of water. After five days the seeds that have sprouted are counted.

Independent variable (and values):

Temperature: 5, 15, 25 and 35 °C.

Dependent variable (and units): **Number of seeds sprouted (out of 20) after five days.**

Three controlled variables: **Any three: 20 seeds per dish; same type of seed; 10 mL of water; same paper towel; identical dishes with lids; five days.**

Control group: **None (accept a justified baseline).**